

CS698B: Fundamentals of Data Engineering (Part I)
2026-27-I, IIT Kanpur – Problem Sheet 2



General Instructions:

1. This problem sheet need not be submitted.
2. The questions are meant for practice and self-assessment.
3. Please use this resource honestly to assess your understanding of concepts.

Q1 (Complete the Code). The completed code should count how many odd numbers are in `nums`

```
nums = [3, 8, 5, 2, 9]
count = 0
for x in nums:
    if ____:
        ____
print(count)
```

Q2 (Become the Judge). The following code is meant to count the number of “bulls” (i.e. the number of locations with the same digit) in the strings `guess` and `secret` (assume both have equal length). The code has a bug. Design a valid test case to reveal the bug where the code gives wrong output.

```
secret = input()
guess = input()
bulls = 0
for i in range( len( secret ) ):
    if secret[ i ] == guess[ i ]:
        bulls = bulls + 1
        break
print( bulls )
```

Q3 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
s = "banana"
total = 0
for i in range( len( s ) - 1, -5, -1 ):
    print( s[ i ] )
    if s[ i ] == 'a':
        total = total + 1
print( total )
```

Q4 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
n = 3
for i in range( 1, n + 1 ):
    line = []
    for j in range( 1, i + 1 ):
        line.append( str( i * j ) )
    print( ' '.join( line ) )
```

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Q5 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
a = [ 1, 2, 3 ]
b = a
c = a[ : ]
a.append( 4 )
print( a )
print( b )
print( c )
```

Q6 (Complete the Code). The completed code so that `unique_lengths` becomes `{3, 5}`

```
words = [ "cat", "dog", "mouse", "ant" ]
unique_lengths = { _____ for w in words }
```

Q7 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
nums = [ 1, 2, 3, 4, 5 ]
parity = { n : "even" if ( n & 1 ) == 0 else "odd" for n in nums }
print( parity[ 3 ] )
print( len( parity ) )
```

Q8 (Become the Judge and Jailor). The following code is meant to count to count how many times each character appears in a string. The code has a bug. Design a valid test case to reveal the bug where the code gives wrong output. What corrections are needed to the code to make it start working properly?

```
text = input()
counts = {}
for ch in text:
    counts[ ch ] = counts.get( ch, 0 )
print( counts )
```

Q9 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
data = ( 1, 2, 2, 3, 3, 3 )
s = { data }
t = tuple( { 1, 2, 2, 3, 3, 3 } )
print( s )
print( t )
```

Q10 (Become the Judge and Jailor). The following code is meant to create a list of squares of even numbers from the list `nums`. The code has a bug. Design a valid test case to reveal the bug where the code gives wrong output. What corrections are needed to the code to make it start working properly?

```
nums = [ 1, 2, 3, 4, 5, 6 ]
squares = [ n * n for n in nums if n % 2 ]
print( squares )
```